

Ganit

data speaks

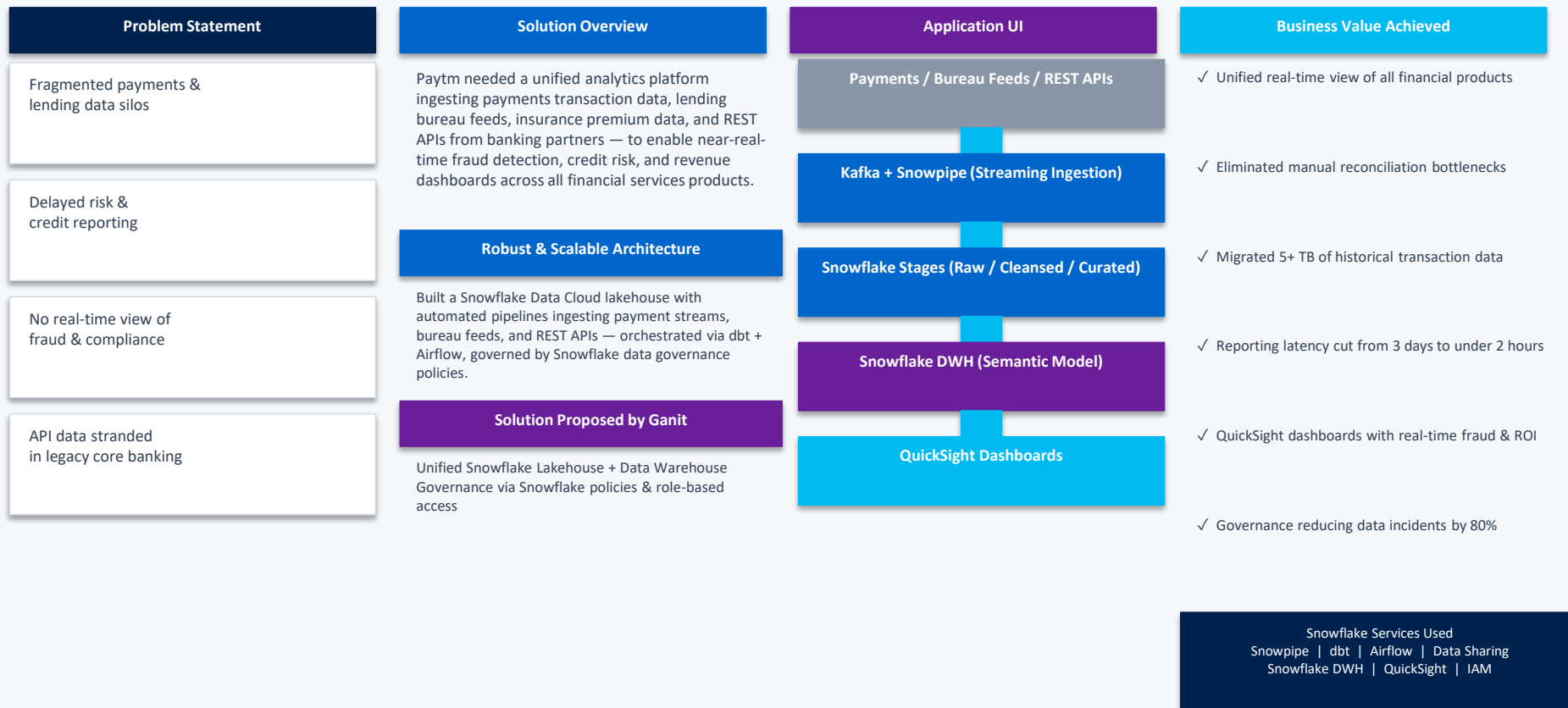
Paytm – Snowflake & QuickSight

Case Study

Transforming Payments, Lending & Insurance Analytics with a Scalable Snowflake Data Platform & QuickSight Frontend

BFSI | Fintech & Financial Services

Modernizing Payments, Lending & Insurance Analytics for India's Leading Fintech with Snowflake – Enabling Real-Time Risk, Fraud & Revenue Dashboards Across Digital Financial Services



Ganit unified Paytm's payments, lending & insurance data on Snowflake, accelerating analytics adoption and enabling real-time fraud, credit risk & revenue decisions across all digital financial services

Outcome

Delivered a cloud-native Snowflake Data Cloud lakehouse ingesting data from payment gateways, Kafka event streams, credit bureau APIs, insurance premium feeds, and banking partner REST APIs — governed by Snowflake role-based access and column-level security — and surfaced via Amazon QuickSight for self-service analytics and automated regulatory reporting across 10+ financial products.

Operational Efficiency

Eliminated manual reconciliation and siloed payment exports by automating ingestion via Kafka + Snowpipe streaming — reducing data latency from 3 days to under 2 hours and freeing analyst time for product analytics and risk modelling.

Cost Optimization

Replaced 4 legacy on-premise data warehouses with a unified Snowflake platform using auto-suspend and multi-cluster warehouses — achieving ~80% reduction in BI licensing and infrastructure costs while enabling elastic scaling for month-end peaks.

Scalability & Reliability

Built a multi-layer Snowflake lakehouse (Raw / Cleansed / Curated) with row-level and column-level security, handling 5+ TB of historical transaction data and supporting 300+ concurrent QuickSight users across business functions.

Business Growth

Enabled real-time fraud detection, credit risk, and revenue performance dashboards in QuickSight, giving risk, product, and finance teams actionable visibility into transaction anomalies, loan book health, and product-level P&L.

Siloed payment streams, manual bureau uploads, and stranded core banking APIs blocked unified financial analytics across Paytm's digital lending, insurance & payments business

Problem Statement

Paytm operates one of India's largest fintech ecosystems spanning UPI payments, buy-now-pay-later, personal loans, insurance distribution, and wealth management — processing hundreds of millions of daily transactions across multiple product lines.

Data was siloed across payment gateways, core banking systems, credit bureau APIs, and insurance partner feeds — with no unified risk or revenue view. Overnight batch jobs created T+3 delays, and compliance teams relied on manual Excel reconciliation for RBI regulatory submissions.

Challenge 1

Fragmented payment gateway & bureau data with no real-time ingestion pipeline

Challenge 2

Manual Excel/CSV reconciliation causing 2–3 day reporting delays for risk & finance

Challenge 3

Core banking & insurance partner REST API data stranded in operational systems

Challenge 4

No unified product-level view of revenue, credit risk & fraud across all verticals

Ganit built a multi-source Snowflake Data Cloud platform with automated ingestion from payment streams, bureau feeds & banking APIs, served through QuickSight for real-time fraud, risk & revenue analytics

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